ASSIGNMENT 02

1. **Find Roots of Equation.**

Code:-

```
#include<iostream>
#include<cmath>
using namespace std;

double firstRoot(int a, int b, int c, int res){
	int d = 2*a;
	double first =(double)(-b+sqrt(res))/d;
	return first;
}

double secondRoot(int a, int b, int c, int res){
	int d = 2*a;
	double second =(double)(-b-sqrt(res))/d;
	return second;
}

int main(){
	int a, b, c;
	cout<<"Enter values for a, b, c: "<<endl;
	cin>>a>>b>>c;
	int res = ((b*b)-(4*a*c));

	if(res > 0)
	{
		cout<<"Two Real Roots."<<endl;
		cout<< firstRoot(a,b,c,res)<<endl;
		cout<< secondRoot(a,b,c,res)<<endl;
	}
	else if(res == 0)
	{
		cout<<"One Real Roots."<<endl;
		cout<< firstRoot(a,b,c,res)<<endl;
		cout<< secondRoot(a,b,c,res)<<endl;
```

```
        }
        else{
                cout<<"Roots are Complex."<<endl;
        }
        return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./rootsofequation.out
Enter values for a, b, c:
1
-7
10
Two Real Roots.
5
2
pgcp-ac@dac:~/Batch2/Assignment 02$
```

2. **Check if given number is odd or even.**

Code:-

```
#include<iostream>
using namespace std;

int main(){
	int num;
	cout<<"Enter number: "<<endl;
	cin>>num;
	if(num==0)
	{
		cout<<"Given number is 0."<<endl;
	}
	else if (num%2==0){
		cout<<"Given number is Even."<<endl;
	}
	else{
		cout<<"Given number is Odd."<<endl;
	}
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./evenodd.out
Enter number:
35
Given number is Odd.
pgcp-ac@dac:~/Batch2/Assignment 02$ ./evenodd.out
Enter number:
-2
Given number is Even.
pgcp-ac@dac:~/Batch2/Assignment 02$
```

3. **Check if the entered char is a small letter or capital letter.**

Code:-

```
#include<iostream>
using namespace std;

int main(){
	char ch;
	cout<<"Enter any character: "<<endl;
	cin>>ch;
	if((ch>=65) && (ch<=90))
	{
		cout<<"The char is UpperCase."<<endl;
	}
	else if((ch>=97) && (ch<=122))
	{
		cout<<"The char is LowerCase."<<endl;
	}
	else{
		cout<<"Invalid Input!!!"<<endl;
	}
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./charactercheck.out
Enter any character:
G
The char is UpperCase.
pgcp-ac@dac:~/Batch2/Assignment 02$ ./charactercheck.out
Enter any character:
s
The char is LowerCase.
pgcp-ac@dac:~/Batch2/Assignment 02$
```

4. **Write a recursive function to obtain the running sum of n numbers.**

Code:-

```
#include<iostream>
using namespace std;

int sum(int no){
	if(no==1)
	{
		return 1;
	}
	return no+sum(no-1);
}

int main(){
	int n, tot_sum=0;
	int ch;
	do{
	cout<<"Enter the value for n: "<<endl;
	cin>>n;

	for(int i=1; i<=n; i++)
	{
		tot_sum += sum(i);
	}
	cout<<"Total sum of running numbers is: "<<tot_sum<<endl;
	cout<<"\n1. Do You Want to Continue.\n2.Exit "<<endl;
	cin>>ch;
	}while(ch==1);
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./runningsumRecursive.out
Enter the value for n:
7
Total sum of running numbers is: 84
1. Do You Want to Continue.
2.Exit
1
Enter the value for n:
10
Total sum of running numbers is: 304
1. Do You Want to Continue.
2.Exit
2
pgcp-ac@dac:~/Batch2/Assignment 02$
```

5. **Write functions for calculating area of circle, rectangle, and square and call these functions from the main function.**

Code:-

```
#include<iostream>
using namespace std;
const double pi = 3.14;

double area_circle(int r){
	double c_area = pi * r * r;
	return c_area;
}

int area_rectangle(int l, int b){
	int r_area = l * b;
	return r_area;
}

int area_square(int s){
	int s_area = s * s;
	return s_area;
}

int main(){

	//Area of circle
	int radius;
	cout<<"Enter radius of a circle: "<<endl;
	cin>>radius;
	cout<<"Area of circle: "<<area_circle(radius)<<endl;

	//Area of Rectangle
	int len, bre;
	cout<<"\nEnter Length and Breadth of Rectangle"<<endl;
	cin>>len>>bre;
	cout<<"Area of rectangle: "<<area_rectangle(len, bre)<<endl;

	//Area of square
	int side;
```

```
        cout<<"Enter the side of square: "<<endl;
        cin>>side;
        cout<<"Area of square: "<<area_square(side)<<endl;
        return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./areas.out
Enter radius of a circle:
5
Area of circle: 78.5

Enter Length and Breadth of Rectangle
30
20
Area of rectangle: 600
Enter the side of square:
50
Area of square: 2500
pgcp-ac@dac:~/Batch2/Assignment 02$
```

6. **Write a function for factorial and use the function for printing factorial series like 1 2 6 24 120.....**

Code:-

```
#include<iostream>
using namespace std;

int factorial(int num){
	if(num==1)
	{
		return 1;
	}
	return num*factorial(num-1);
}

int main(){
	int k;
	cout<<"Enter value of K: "<<endl;
	cin>>k;
	if(k==0)
	{
		cout<<"0"<<endl;
	}
	for(int i=1; i<=k; i++)
	{
		cout<<factorial(i)<<endl;
	}
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 02$ ./factseries.out
Enter value of K:
7
1
2
6
24
120
720
5040
pgcp-ac@dac:~/Batch2/Assignment 02$
```